

Advanced Blockchain Security Framework — Documentation

AI-DRIVEN ADVANCED BLOCKCHAIN SECURITY FRAMEWORK — DETAILED DOCUMENTATION

1. INTRODUCTION

The AI-Driven Advanced Blockchain Security Framework (ABSF) merges machine learning anomaly detection with zero-trust node authentication and secure smart contract design. It defends against fraudulent transactions, malicious nodes, and contract exploits.

2. SYSTEM ARCHITECTURE

The framework has five layers:

- Blockchain Core – transaction pool, blocks, mining, consensus.
- AI Security Agent – IsolationForest-based anomaly scoring.
- Zero-Trust Policy – JWT authentication for nodes.
- Smart-Contract Layer – secure Solidity examples.
- Visualization Dashboard – Streamlit-based analytics.

3. COMPONENT DESIGN

- Blockchain Simulator: Implements block mining, hashing, and PoW with a security hook that filters suspicious transactions.
- AI Anomaly Detection: IsolationForest model trained on synthetic blockchain data to detect anomalies.
- Zero-Trust Authentication: JWT-based access control, rejecting unauthorized node interactions.
- Smart Contracts: "VulnerableBank.sol" (reentrancy issue) and "SecureWallet.sol" (protected with reentrancy guard).
- Dashboard: Streamlit dashboard visualizing anomaly scores, suspicious transactions, and score distributions.

4. MACHINE LEARNING MODEL

Uses Scikit-learn IsolationForest trained on features: amount, frequency per hour, gas price, contract call, hour. Anomalous transactions exceed the threshold (score > 0.2).

5. API DOCUMENTATION

Endpoints:

- GET /health → API status check
- POST /tx → Submit transaction (requires JWT)
- POST /mine → Mine filtered block (requires JWT)
- GET /chain → Retrieve blockchain state

6. DEPLOYMENT GUIDE

Local setup:

```
> pip install -r backend/requirements.txt
> uvicorn backend.api.app:app --reload
> streamlit run dashboard/streamlit_app.py
```

Docker:

```
> docker compose up --build
API: http://localhost:8000
Dashboard: http://localhost:8501
```

7. SECURITY MECHANISMS

Layer	Technique	Purpose
Transaction	ML anomaly detection	Fraud prevention

Node	Zero-trust JWT	Node-level auth
Contract	Checks-Effects-Interactions	Reentrancy prevention
Analytics	Dashboard	Real-time threat visualization

8. SAMPLE RESULTS

Detection Accuracy: >95%

Top 20 Suspicious TX visualized in dashboard

Histogram of anomaly scores displayed

9. FUTURE ENHANCEMENTS

- Federated AI for multi-chain threat sharing
- GNN for blockchain relationship analysis
- Homomorphic encryption for privacy-preserving ML
- LLMs for automatic smart-contract auditing
- Cross-chain zero-trust federation

10. REFERENCES

- Scikit-learn IsolationForest
- OpenZeppelin ReentrancyGuard
- NIST SP 800-207 Zero Trust Architecture
- Ethereum Smart Contract Best Practices
- OWASP Blockchain Security Guidelines